Automated Data Testing with the Snowflake Semantic Layer

This document explains the end-to-end example built in this session: a semantic view on TPC-H sample data with verified queries used as automated test assertions, scheduled to run daily.

Overview

The semantic layer in Snowflake (via **Semantic Views**) provides a business-friendly abstraction over raw tables. **Verified Queries (VQRs)** are known-good SQL paired with natural language questions that guide Cortex Analyst. This example repurposes VQRs as regression tests — if the underlying data changes shape, the tests catch it.

What Was Created

All objects live in `SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS`.

OBJECT	TYPE	PURPOSE
`TPCH_REVENUE_SV`	Semantic View	Business model over TPC-H data
`TEST_SEMANTIC_LAYER()`	Procedure	Runs 6 automated tests, returns pass/fail table
`RUN_AND_LOG_TESTS()`	Procedure	Wrapper that logs results to history
`TEST_HISTORY`	Table	Persists all test runs for trending
`DAILY_SEMANTIC_TESTS`	Task	Runs tests daily at 6 AM ET

The Semantic View

Built on `SNOWFLAKE_SAMPLE_DATA.TPCH_SF1` with:

- **5 tables**: REGION, NATION, CUSTOMER, ORDERS, LINEITEM
- **4 relationships**: region -> nation -> customer -> orders -> lineitem
- **4 facts**: order_total, extended_price, discount, quantity
- **7 dimensions**: region_name, nation_name, customer_name, market_segment, order_date, order_status, ship_date
- **3 metrics**: total_revenue, order_count, avg_order_value
- **3 verified queries** (see below)

Verified Queries as Test Assertions

Each VQR pairs a question with its expected SQL:

VQR NAME	QUESTION	ASSERTION
`revenue_by_region`	"What is the total revenue by region?"	Returns exactly 5 rows
`top_customers_by_spend`	"Who are the top 10 customers by total spend?"	Returns exactly 10 rows
`avg_order_by_segment`	"What is the average order value by market segment?"	Returns exactly 5 rows

The 6 Automated Tests

```
TEST_NAME                   | What It Checks
---------------------------|---------------------------------------
---------
semantic_view_exists        | The semantic view is deployed and
accessible
vqr_revenue_by_region       | VQR SQL runs and returns 5 regions
vqr_top_customers           | VQR SQL runs and returns 10 customers
vqr_avg_order_by_segment    | VQR SQL runs and returns 5 segments
source_data_present         | Underlying LINEITEM table has rows
metric_revenue_positive     | Revenue metric is positive for all
regions
```

How to Run

```sql
-- Run tests interactively (returns a table)
CALL SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS.TEST_SEMANTIC_LAYER();

-- Run and persist results to history
CALL SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS.RUN_AND_LOG_TESTS();

-- View historical results
SELECT run_at, test_name, status, details
FROM SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS.TEST_HISTORY
ORDER BY run_at DESC, test_name;

-- Check task status
SHOW TASKS IN SCHEMA SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS;
```

Architecture

```
+---------------------+       +-------------------------+
|  TPCH_REVENUE_SV    |       |  TEST_SEMANTIC_LAYER()  |
|  (Semantic View)    |       |  (Test Procedure)       |
|                     |       |                         |
|  - Tables           |       |  - Existence check      |
|  - Relationships    |<------+  - VQR SQL execution    |
|  - Metrics          |       |  - Data freshness       |
|  - Verified Queries |       |  - Metric sanity        |
+---------------------+       +-------------------------+
                                         |
                                         v
                              +-------------------------+
                              |  RUN_AND_LOG_TESTS()    |
                              |  (Logging Wrapper)      |
                              +-------------------------+
                                         |
                                         v
+---------------------+       +-------------------------+
|  TEST_HISTORY       |<------|  DAILY_SEMANTIC_TESTS   |
|  (Results Table)    |       |  (Scheduled Task)       |
```

```
|                      |        |  Cron: 0 6 * * *         |
+----------------------+        +-------------------------+
```

Key Design Decisions

1. **VQRs serve dual purpose** — They guide Cortex Analyst for user queries AND act as regression tests. One artifact, two benefits.
2. **Tests are idempotent** — Each test is a SELECT that checks row counts or value constraints. No side effects on source data.
3. **History enables trending** — The `TEST_HISTORY` table lets you detect when a test started failing and correlate with upstream changes.
4. **Procedure returns a table** — `TEST_SEMANTIC_LAYER()` returns structured results, so it can be consumed by dashboards or alerting.

Extending This Pattern

- **Add more VQRs**: Each new verified query is both a Cortex Analyst hint and a new test case.
- **Add value assertions**: Check that `total_revenue` for EUROPE is within an expected range (not just positive).
- **Use `EXECUTE_AI_EVALUATION`**: Run Cortex Analyst evaluations to test whether the AI *generates* correct SQL from natural language, not just that the VQR SQL compiles.
- **Add alerting**: Create an Alert that fires when any test returns FAIL status.
- **Cross-environment testing**: Run the same tests against dev/staging/prod semantic views to catch drift.

Cleanup

To remove everything:

```
ALTER TASK
SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS.DAILY_SEMANTIC_TESTS SUSPEND;
DROP SCHEMA SNOWFLAKE_LEARNING_DB.SEMANTIC_TESTS CASCADE;
```